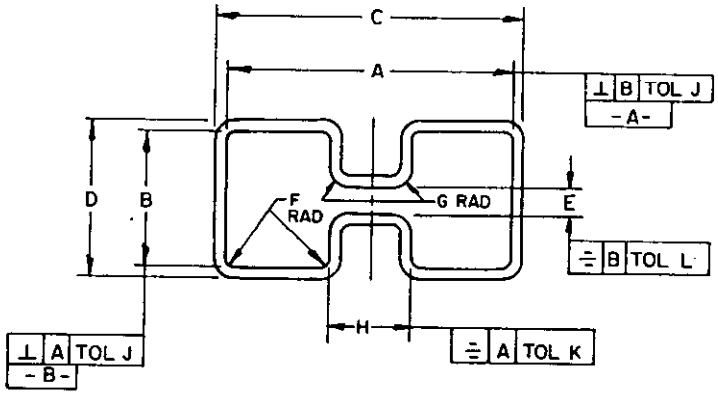
MILITARY SPECIFICATION SHEET

WAVEGUIDES, DOUBLE RIDGE

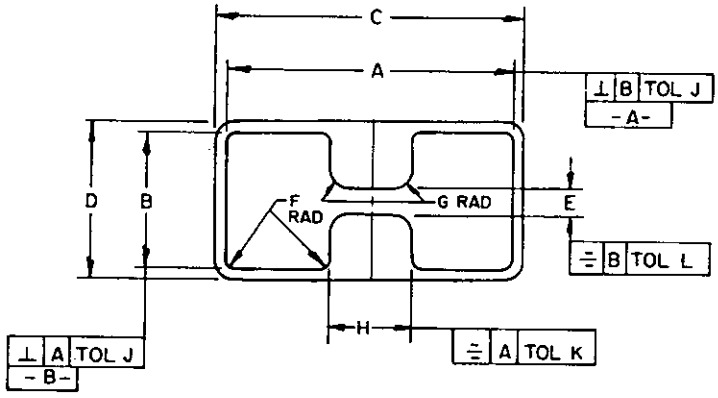
(BANDWIDTH RATIO 3.6:1)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the waveguides described herein shall consist of this document and the latest issue of Specification MIL-W-23351.



Configuration 1.



Configuration 2.

FIGURE 1. Waveguide.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, Department of the Navy, ATTN: ELEX 5043, Washington, D.C. 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5985

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TABLE I. Part number dimensions and material.

Part number M23351/2-	Dimensions 1/				
	A	B	C	D	E
001 2/ 002 003 004	34.638 ±.020 (879.80) (.51)	14.894 ±.020 (378.31) (.51)	34.888 Max (886.16)	15.144 Max (384.66)	2.904 ±.005 (73.76) (.13)
005 006 007 008	13.916 ±.010 (353.47) (.25)	5.984 ±.010 (152.00) (.25)	14.166 Max (359.82)	6.234 Max (158.34)	1.167 ±.003 (29.64) (.08)
009 010 011 012	9.628 ±.010 (244.55) (.25)	4.140 ±.010 (105.16) (.25)	9.878 ±.015 (250.90) (.38)	4.390 ±.015 (111.51) (.38)	.8070 ±.003 (20.498) (.08)
013 014 015 016	3.877 ±.004 (98.48) (.10)	1.667 ±.004 (42.34) (.10)	4.037 ±.005 (102.54) (.13)	1.827 ±.005 (46.41) (.13)	.3250 ±.003 (8.255) (.08)
017 018 019 020 2/	2.687 ±.004 (68.25) (.10)	1.155 ±.004 (29.34) (.10)	2.847 ±.004 (72.31) (.10)	1.315 ±.004 (33.40) (.10)	.225 ±.002 (5.72) (.05)
021 3/ 022 023 024	1.074 ±.003 (27.28) (.08)	.462 ±.003 (11.73) (.08)	1.174 ±.004 (29.82) (.10)	.562 ±.004 (14.27) (.10)	.090 ±.002 (2.29) (.05)
025 026 027 028	.752 ±.003 (19.10) (.08)	.323 ±.003 (8.20) (.08)	.852 ±.004 (21.64) (.10)	.423 ±.004 (10.74) (.10)	.063 ±.002 (1.60) (.05)
029 030 031 032 3/	.303 ±.002 (7.70) (.05)	.130 ±.002 (3.30) (.05)	.383 ±.003 (9.73) (.08)	.210 ±.003 (5.33) (.08)	.025 ±.002 (.64) (.05)

See footnotes at end of table.

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TABLE I. Part number dimensions and material - Continued.

Part number M23351/2-	Dimensions (continued) 1/						
	F Max	G ±10%	H	J	K	L	Material
001 2/ 002 003 004	.067 (1.70)	.581 (14.76)	8.660 ±.010 (219.96) (.25)	.020 (.51)	.010 (.25)	.005 (.13)	Aluminum alloy Copper alloy Copper Silver alloy
005 006 007 008	.055 (1.40)	.233 (5.92)	3.479 ±.005 (88.37) (.13)	.010 (.25)	.005 (.13)	.003 (.08)	Aluminum alloy Copper alloy Copper Silver alloy
009 010 011 012	.050 (1.27)	.161 (4.09)	2.407 ±.004 (61.14) (.10)	.010 (.25)	.004 (.10)	.003 (.08)	Aluminum alloy Copper alloy Copper Silver alloy
013 014 015 016	.050 (1.27)	.065 (1.65)	.9690 ±.003 (24.613) (.08)	.004 (.10)	.003 (.08)	.003 (.08)	Aluminum alloy Copper alloy Copper Silver alloy
017 018 019 020 2/	.050 (1.27)	.045 (1.14)	.672 ±.002 (17.07) (.05)	.004 (.10)	.002 (.05)	.002 (.05)	Aluminum alloy Copper alloy Copper Silver alloy
021 3/ 022 023 024	.030 (.76)	.018 (.46)	.269 ±.002 (6.83) (.05)	.003 (.08)	.002 (.05)	.002 (.05)	Aluminum alloy Copper alloy Copper Silver alloy
025 026 027 028	.020 (.51)	.013 (.33)	.188 ±.002 (4.78) (.05)	.003 (.08)	.002 (.05)	.002 (.05)	Aluminum alloy Copper alloy Copper Silver alloy
029 030 031 032 3/	.015 (.38)	.005 (.13)	.076 ±.002 (1.93) (.05)	.002 (.05)	.002 (.05)	.002 (.05)	Aluminum alloy Copper alloy Copper Silver alloy

1/ Dimensions are in inches. Metric equivalents (in parentheses) are given for general information only and are based upon 1.00 inch = 25.4 mm.
2/ The waveguide shall be in accordance with configuration 1.
3/ The waveguide shall be in accordance with configuration 2.

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REQUIREMENTS:

Design, construction, and physical dimensions: See figure 1 and table I.
Material: See table I.
Part number: M23351/2- (dash number from table I).

NOTES:

Engineering information and frequency range: See table II.
Cross reference: See table III.

TABLE II. Engineering information.

Part number M23351/2-	Suggested frequency range TE_{10} mode	Cutoff frequency		Theoretical attenuation at $f = \sqrt{3}\, f_{c_{10}}$ 1/ 2/	Theoretical power handling capacity at $f = \sqrt{3}\, f_{c_{10}}$ 3/
		TE_{10} mode	TE_{20} mode		
	GHz	GHz	GHz	dB/ft	kW
001-XXX	.108 - .390	.092	.401	.00065	28,830
002-XXX				.00064	
003-XXX				.00043	
004-XXX				.00046	
005-XXX	.270 - .970	.229	.999	.00235	4,653
006-XXX				.00232	
007-XXX				.00156	
008-XXX				.00167	
009-XXX	.390 - 1.40	.331	1.444	.00415	2,227
010-XXX				.00410	
011-XXX				.00276	
012-XXX				.00296	
013-XXX	.970 - 3.50	.822	3.587	.0155	361.2
014-XXX				.0153	
015-XXX				.0103	
016-XXX				.0110	
017-XXX	1.40 - 5.00	1.186	5.176	.0283	173.5
018-XXX				.0279	
019-XXX				.0188	
020-XXX				.0201	
021-XXX	3.50 - 12.40	2.966	12.944	.110	27.74
022-XXX				.108	
023-XXX				.0729	
024-XXX				.0781	
025-XXX	5.00 - 18.00	4.237	18.490	.192	13.59
026-XXX				.190	
027-XXX				.128	
028-XXX				.137	
029-XXX	12.40 - 40.00	10.508	45.857	.737	2.210
030-XXX				.727	
031-XXX				.490	
032-XXX				.525	

1/ $f_{c_{10}}$ = cutoff frequency for TE_{10} mode.

2/ Actual attenuation of waveguides may be considerably greater depending upon operating frequency and temperature. For further information or to calculate for frequencies other than those tabulated, see S. Hopfer, "The Design of Ridged Waveguides", IRE Transactions on Microwave Theory and Techniques, MTT-3 pp 20-29, October 1955, and MIL-HDBK-216, "Transmission Lines and Fittings", chapter 6. Typical resistivities of materials (at 20° C): aluminum alloy (6061) = 4.0 microhms-cm; Copper alloy (brass) = 3.9 microhm-cm; copper = 1.77 microhm-cm; silver alloy (coin silver) = 2.03 microhm-cm.

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3/ Actual power handling capability of waveguides may be considerably less depending upon waveguide temperature, altitude, operating frequency, etc. For further information or to calculate values for frequencies other than those tabulated, see S. Hopfer, "The Design of Ridged Waveguides", IRE Transactions on Microwave Theory and Techniques, MTT-3 pp 20-29, October 1955, and MIL-HDBK-216, "Transmission Lines and Fittings", chapter 6. The tabulated values were calculated based upon the E field produced breakdown in a nonpressurized air dielectric waveguide under continuous wave (CW) conditions. The breakdown strength of air was considered to be 15000 volts per centimeter, corner radii considered.

TABLE III. Cross reference (type designation to part number).

Type designation	Part number M23351/2-	Type designation	Part number M23351/2-
WRD108U36A	001	WRD140D36A	017
WRD108U36B	002	WRD140D36B	018
WRD108U36C	003	WRD140D36C	019
WRD108U36S	004	WRD140D36S	020
WRD270U36A	005	WRD350D36A	021
WRD270U36B	006	WRD350D36B	022
WRD270U36C	007	WRD350D36C	023
WRD270U36S	008	WRD350D36S	024
WRD390U36A	009	WRD500D36A	025
WRD390U36B	010	WRD500D36B	026
WRD390U36C	011	WRD500D36C	027
WRD390U36S	012	WRD500D36S	028
WRD970U36A	013	WRD124C36A	029
WRD970U36B	014	WRD124C36B	030
WRD970U36C	015	WRD124C36C	031
WRD970U36S	016	WRD124C36S	032

Custodian:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army -
Navy - OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AR
Navy - MC, AS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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